Product Description

Product Information	
Product name	FLIM Data Acquisition Card
Product number	FL-DAQ-02 FL-DAQ-04 FL-DAQ-08
Manufacturer	FLIM LABS S.r.l.
Registered Address	Via della Farnesina, 3 – 00135 Rome (RM), Italy



General Description

The FLIM Data Acquisition Card (DAQ) is a compact data acquisition module designed for the acquisition, processing, and transmission of high-frequency digital electronic signals. It is intended for applications such as time-correlated measurements, event counting, and fluorescence lifetime imaging (FLIM) in scientific and laboratory instrumentation where high timing precision, reliability, and integration with external detection systems are required.

The device is designed to receive high-frequency digital input signals in LVTTL format (50 Ω) from external devices such as single-photon detectors, timing modules, or signal-conditioning electronics. Acquired signals are processed internally and transferred to a host system for analysis and data processing.

In addition to the detector input channels, the device provides dedicated synchronization inputs for Pixel Clock, Line Clock, and Frame Clock signals. These inputs allow the DAQ to correlate detected events with the position of a scanning system, enabling applications such as fluorescence lifetime imaging (FLIM), laser scanning microscopy, and other time-resolved imaging techniques.

The operating principle is based on programmable digital logic (FPGA), enabling flexible and deterministic processing of digital signals and precise management of timing events. The electronic architecture supports repetition frequencies up to 80 MHz, enabling high temporal resolution measurements. The device provides dedicated interfaces for synchronization, including high-frequency input trigger ports from external pulsed laser sources, trigger output ports for synchronizing external lasers or timing devices, and imaging synchronization inputs for pixel, line, and frame timing signals.

Communication with the host system is performed via USB, which is used both for data transfer and for device control and configuration. Power is supplied via 5 V DC directly from the USB interface.

The product is available in multiple variants that differ exclusively in the number of available input and output ports, while maintaining the same operating principle, technology, and safety characteristics.
The module features:

- FPGA-based digital processing for deterministic timing and event handling
- High-frequency detector inputs in LVTTL format (50 Ω) via SMA and LVDS via USB Type-C
- Dedicated Pixel Clock, Line Clock, and Frame Clock synchronization inputs for imaging applications
- Dedicated laser trigger input and laser trigger output for synchronization
- USB interface for configuration and data transfer (also providing 5 V power)
- Compact plastic enclosure
- Variant-dependent number of detector input channels (DAQ-02 / DAQ-04 / DAQ-08)

Intended Use and Misuse

Intended Use

The FLIM Data Acquisition Card is intended as a compact electronic module for acquisition, processing, and transmission of electronic signals by professional users in private research laboratories, industrial environments, or universities. It may be used as a stand-alone unit or integrated into custom measurement systems and scientific instrumentation.

Reasonably Foreseeable Misuse

The following uses are not permitted:

- Use outside described applications
- Use with incompatible components
- Operation outside specified limits
- Unauthorized modification
- Use with incorrect power supply

Environmental and Operating Conditions

The device is designed for indoor laboratory use only. Operate the device only within the specified environmental limits:

- Operating temperature: 0°C to +40°C
- Storage temperature: −20°C to +70°C
- Relative humidity: up to 80% non-condensing
- Stable, vibration-free surface

Do not operate the device in explosive atmospheres or in the presence of flammable gases.
Ensure adequate ambient lighting during installation and adjustment to avoid connection errors.